Mapping Philosophy of Science at Scale with the SOPHIS Database

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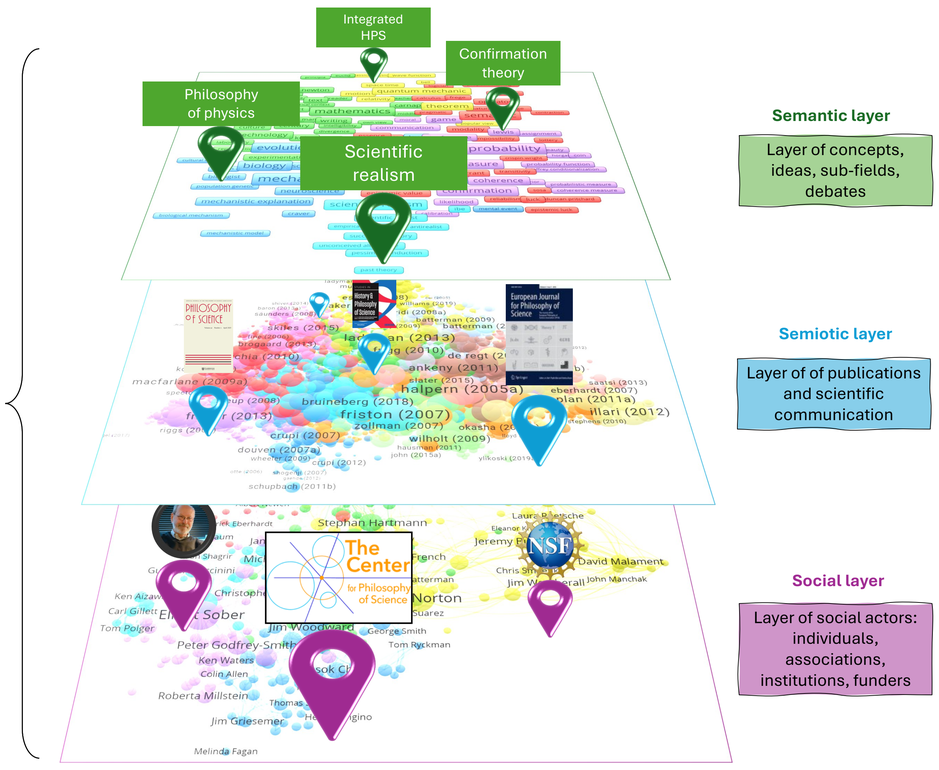
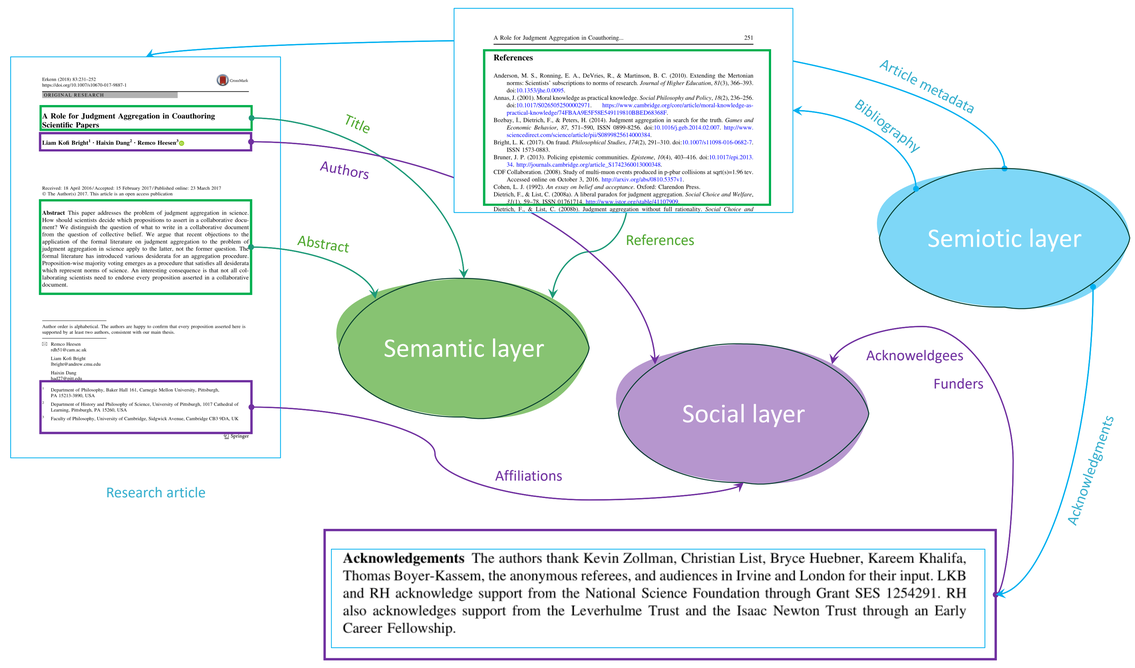
The structure of philosophy of science

1 We model philosophy of science as a **structure** made of **three interrelated layers**: the **semantic**, **semiotic**, and **social**



2 We can use **information** in **publications** **(the semiotic layer)** to **map** the **semantic** and **social** layers of philosophy of science



3 The **acknowledgements** in academic publications, in particular, provide precious data on **informal collaborations** and **funding organizations**, allowing to uncover the **«hidden part» of the social layer**

The SOPHIS database

SOPHIS (**S**ocial **O**bservatory of **PHI**losophy of **S**cience) is a **graph database** including **multi-dimensional data** on the **semantic, semiotic, and social layer of philosophy of science** extracted from almost 7,000 philosophy of science publications

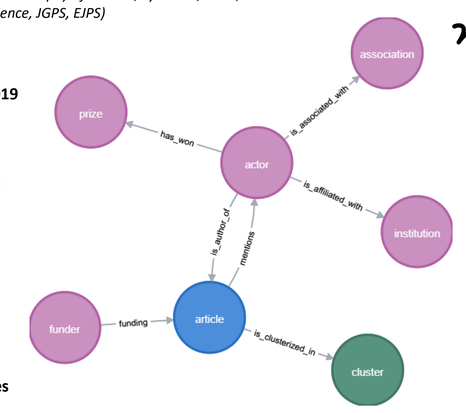
8 journals (*Erkenntnis, Philosophy of Science, Synthese, SHPS, BJPS, Foundations of Science, JGPS, EJPS)*

6,826 resarch articles from **2005 to 2019**

10,876 individual actors (**4,380 authors** and **8,952 acknowledgees**)

2,105 institutions from **77 countries**

515 funders from **51 countries**



SOPHIS meta-graph

5,376 acknowledgments manually collected (**79%** of articles include acknowledgements)

46 clusters of publications obtained combining advanced citation analysis with artificial intelligence

200 officials of **6 professional associations** (PSA, EPSA, &HPS, SPSP, SRPoiSE, SMS)

60 winners of **3 prizes** (Lakatos Award, Hempel Award, Popper Award)

Mapping philosophy of science

Most mentioned acknowledgees

Actor (Aff. Country)	Mentions	Association(s)	Award(s)
Elliott Sober (USA)	104	PSA*	Lakatos, Hempel
John Norton (USA)	97	&HPS, PSA	
Carl Craver (USA)	66		
Anjan Chakravartty (CAN)	65	SRPoiSE, PSA	
Stephan Hartmann (DEU)	62	EPSA*, PSA	
David Chalmers (AUS)	61		
Alan Hajek (AUS)	60	PSA	
James Woodward (USA)	57	PSA	Lakatos
Branden Fitelson (USA)	55		
Jeremy Butterfield (UK)	54		

Top countries by number of funded articles

Country	Funded articles
United States	432
United Kingdom	306
Germany	271
European Union	172
Canada	149
Spain	130
The Netherlands	111
Belgium	98
Australia	80
Switzerland	55

Factors affecting the number of mentions received

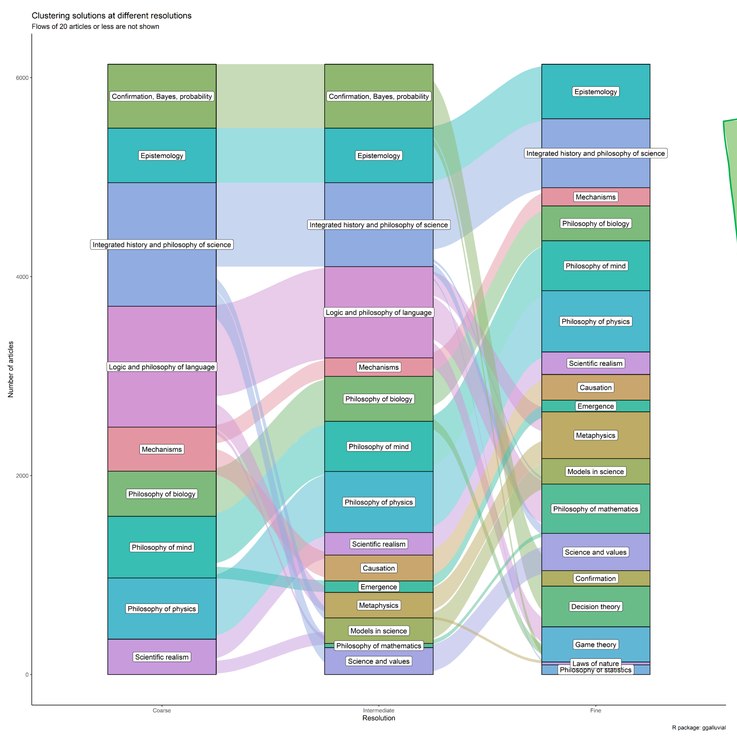
Variable	Estimate (St. Error)
Number of awards received	10.10 (0.57) ***
Number of phil. sci. publications	1.56 (0.034) ***
Affiliation with English-speaking university	1.38 (0.13) ***
Number of governing roles in associations	1.12 (0.12) ***
Gender (1 = male)	0.54 (0.16) ***
Overal publication count	0.006 (0.0016) ***
Citation count	−0.000044 (0.000016) **

OLS regression results for mentions ($N = 5206$) Adjusted R^2= 0.405***

Gender statistics

	All actors	Authors	Acknowledgees	Top mentioned
♂	77.1%	78.8%	77.4%	85.7%
♀	21.7%	19.3%	21.9%	14.3%

Philosophy of science specialties at different levels of granularity



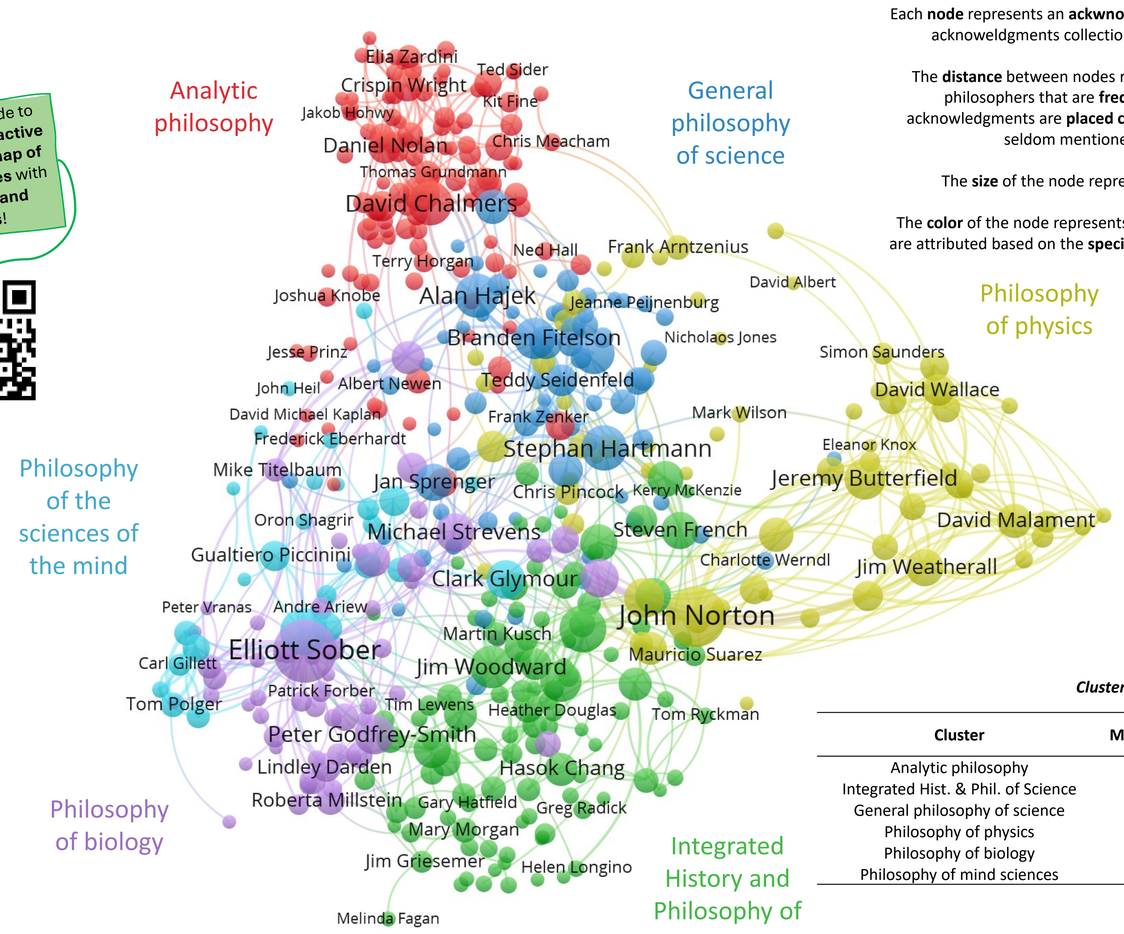
Scan the QR-code to access the **interactive version** of the **map of research articles** with **further data and statistics**!

The VOSviewer **clustering algorithm** is used to identify **groups of articles that share many references** (bibliographic coupling similarity). The algorithm is governed by a **resolution parameter**: the higher the parameter, the more clusters are identified.

The diagram shows **relationships between clusters** at three levels of resolution. Flows represent how clusters at one resolution level distribute their articles across clusters at the next level.

Cluster **labels** are assigned based on a **summary** of the abstracts of the articles in each cluster, generated using **artificial intelligence** (KeyBERT).

Social map of philosophy of science



Each **node** represents an **ackwnoweldgee** mentioned at least 10 times in SOPHIS' acknowledgments collection ($n = 447$, top 5% of all acknowledgees).

The **distance** between nodes represent the **co-acknowledgment similarity**: philosophers that are **frequently mentioned together** in the same acknowledgments are **placed closer** on the map, while philosophers that are seldom mentioned together are placed far apart.

The **size** of the node represents the **number of mentions** received.

The **color** of the node represents the **cluster** the node belongs to. Cluster labels are attributed based on the **specialization** of the acknowledgees belonging to the cluster.

Scan the QR-code to access the **interactive version** of the social map with **further data** and **statistics**!

Cluster statistics

Cluster	Members	Avg. Mentions	Awards	Associations officials (%)
Analytic philosophy	117	16.5	0	0 (0%)
Integrated Hist. & Phil. of Science	108	20.0	2	50 (46.3%)
General philosophy of science	71	19.4	14	9 (12.7%)
Philosophy of physics	66	22.3	12	15 (22.7%)
Philosophy of biology	58	21.6	12	7 (12.1%)
Philosophy of mind sciences	27	21.8	1	4 (14.8%)

Research project PHIL-IMPACT || Funded by PNRR and European Union